In an examination, a student can choose the order in which two questions (QuesA and QuesB) must be attempted.

- If the first question is answered wrong, the student gets zero marks.
- If the first question is answered correctly and the second question is not answered correctly, the student gets the marks only for the first question.
- If both the questions are answered correctly, the student gets the sum of the marks of the two questions.

The following table shows the probability of correctly answering a question and the marks of the question respectively.

question	probabilty of answering correctly	marks
QuesA	0.8	10
QuesB	0.5	20

Assuming that the student always wants to maximize her expected marks in the examination, in which order should she attempt the questions and what is the expected marks for that order (assume that the questions are independent)?

A. First QuesA and then QuesB. Expected marks 14.
B. First QuesB and then QuesA. Expected marks 14.
C. First QuesB and then QuesA. Expected marks 22.
D. First QuesA and then QuesB. Expected marks 16.

As an advertising campaign, a chocolate factory places golden tickets in some of its candy bars, with the promise that a golden ticket is worth a trip through the chocolate factory, and all the chocolate you can eat for life. If the probability of finding a golden ticket is p, find the mean and the variance of the number of candy bars you need to eat to find a ticket.

Two coins are simultaneously tossed until one of them comes up a head and the other a tail. The first coin comes up a head with probability p and the second with probability q. All tosses are assumed independent.

(a) Find the PMF, the expected value, and the variance of the number of tosses.

(b) What is the probability that the last toss of the first coin is a head?

H/W:-

	$E(X)$
HH	6
TT	6
HH or TT	3
HT or TH	3
HT	4
TH	4

An unbiased coin is tossed repeatedly until the outcome of two successive tosses is the same. Assuming that the trials are independent, the expected number of tosses is

A. 3
B. 4
C. 5
D. 6

Ram has a fair coin, i.e., a toss of the coin results in either head or tail and each event happens with probability exactly half $(1/2)$. He repeatedly tosses the coin until he gets heads in two consecutive tosses. The expected number of coin tosses that Ram does is.

A. 2
B. 4
C. 6
D. 8
E. None of the above

When a coin is tossed, the probability of getting a Head is $p, 0 < p < 1$. Let N be the random variable denoting the number of tosses till the first Head appears, including the toss where the Head appears. Assuming that successive tosses are independent, the expected value of N is

A. $\frac{1}{p}$
B. $\frac{1}{(1-p)}$
C. $\frac{1}{p^2}$
D. $\frac{1}{(1-p^2)}$

The joint probability mass function of X and Y is given by

$$\begin{aligned} p(1,1) &= 1/8 & p(1,2) &= 1/4 \\ p(2,1) &= 1/8 & p(2,2) &= 1/2 \end{aligned}$$

(a) Compute the conditional mass function of X given $Y = i$ for $i \in \{1,2\}$.
(b) Are X and Y independent?
(c) Compute $\mathbb{P}(XY < 3)$, $\mathbb{P}(X + Y > 2)$, and $\mathbb{P}(X/Y > 1)$.

Clark likes to buy a certain kind of lottery ticket. Each ticket, independently of all other tickets, yields one of the following prizes.

- \$10, with probability 1/20
- \$1, with probability 1/5
- 2 free lottery tickets, with probability 1/4
- nothing, with probability 1/2

What is the expected payoff of a ticket (in dollars)?

(1) Let X, Z are two R.V, $Y = X \oplus Z$ 　　X, Z are indep.

Find $P(Y=y)$

	$X=0$	$X=1$
$P(X=x)$	p	$1-p$

	$Z=0$	$Z=1$
$P(Z=z)$	ε	$1-\varepsilon$

	$X=0$	$X=1$
$P(X=x)$	p	$1-p$

	$Z=0$	$Z=1$
$P(Z=z)$	ϵ	$1-\epsilon$

Function of R.V

$P(Y=y)$

(1) What all possible values it can take?

$Y = X \oplus Z$ 　　Y is dependent on X, Z

X	Z	$Y = X \oplus Z$
0	0	0
0	1	1
1	0	1
1	1	0

2^2

$Y = X \oplus Z$

$1 \oplus 1 = 0$

$1 \oplus 0 =$

$1 \% 2 = 1$

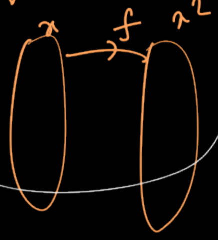
$\oplus \equiv \bmod 2$

$1 \oplus 1 = 2\%2 = 0$

$\therefore$ Y can take 0, 1 values.

Y is R.V., function

$f(x) = x^2$

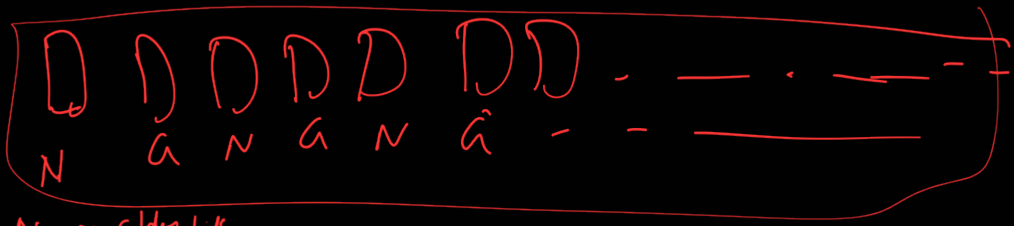


$$P(Y=0) = P((X=0 \cap Z=0) \cup (X=1 \cap Z=1))$$
$$= P(X=0 \cap Z=0) + P(X=1 \cap Z=1)$$
$$= P(X=0)\,P(Z=0) + P(X=1)\,P(Z=1)$$
$$= P(\varepsilon) + (1-P)(1-\varepsilon)$$

$$P(Y=1) = P(X=0)\,P(Y=1) + P(X=1)\,P(Y=0)$$
$$=$$

Y	0	1
$P(Y=y)$	$P\varepsilon + (1-P)(1-\varepsilon)$	$\cdots$

As an advertising campaign, a chocolate factory places golden tickets in some of its candy bars, with the promise that a golden ticket is worth a trip through the chocolate factory, and all the chocolate you can eat for life. If the probability of finding a golden ticket is p, find the mean and the variance of the number of candy bars you need to eat to find a ticket.



N → no golden ticket

G → Golden ticket

Geometric distribution

success → getting an golden ticket.

experiment: picking candies until u find golden ticket

X = # bars u need to eat to find ticket

$\Omega = \{ G, (N,G), (N,N,G), (N,N,N,G), \ldots \}$

set of outcomes

$\boxed{(x,y)} \neq (y,x)$

ordered pair (order matters)

unordered pair: $\{x,y\} = \{y,x\}$

$X \sim geo(p)$

$E[X] = (1/p)$

$$\boxed{Var(X) = \frac{1-p}{p^2}}$$

3 marks (2Q)

(a) 1.5

(b) 1.5

$P(X \geq Y)$

$\sum P(X=K)$

$P(Y \leq K$

Round 1: HT or TH

$X=2$

(HH, HT) (HH, TH)

(TT, HT) (TT, TH)

Two coins are simultaneously tossed until one of them comes up a head and the other a tail. The first coin comes up a head with probability p and the second with probability q. All tosses are assumed independent. $P(A \cap B) = P(A)P(B)$ $P(A|B) = P(A)$

(a) Find the PMF, the expected value, and the variance of the number of tosses.

(b) What is the probability that the last toss of the first coin is a head? A, B are ind

(2 coins there)

C1 (Coin1) C2 (Coin2)

capital P

$P(A) =$ $\boxed{H/T}$

Ω → one Prob P

$P_1(\{H\}) = p$ $\quad P_2(\{H\}) = q$

$P_1(\{T\}) = 1-p$ $\quad P_2(\{T\}) = q$

(T, H) (H, T)
(H, H) (T, T)

P(getting head on coin1) = p

Round 1 : simultaneously tossing 2 coins → All possible outcomes in Round 1
{HH, HT, TH, TT}

Round 2 :

inter & intra

Round1 Round 2

independent

&

In same rounds, coins are independent.

$P_X(x) = P(X = x)$

$X \to$ # of trials until u get success.

X → # of rounds until we stop.

1st round : HT or TH

$X = 1, 2, 3, 4, 5 \ldots$

$P(X=1) = P((HT) \cup (TH)) = P(HT) + P(TH)$

$= P(C_1 = H \cap C_2 = T) + P(C_1 = T \cap C_2 = H)$

$= p(1-q) + (1-p)q$

$= p + q - 2pq$

call as δ.

$P(\text{in one round u stopped}) = \delta$

$P(X=2)$ 1st round 2nd round

F stopped

$P(X=3)$ 1st 2nd 3rd

Fail Fail Stop

$$\boxed{X \sim geo(\delta)}$$

$$P(X=k) = (1-\delta)^{k-1}\delta$$

$$E[X] = 1/\delta$$

$$Var(X) = \left(\frac{1-\delta}{\delta^2}\right)$$

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conditional prob.

last toss $= \{HT, TH\}$ why not HH, TT?

succ $\rightarrow$ stopping $\rightarrow$ HT or TH

$P(\text{first coin is a head} \mid \text{last toss})$

$$P(HT \mid HT \cup TH) =$$

$$= \frac{P((HT) \cap (HT \cup TH))}{P(HT \cup TH)} \qquad HT \cap TH = \phi$$

$A \cap (B \cup C)$

$(A \cap B) \cup (A \cap C)$

$$\cancel{P(HT \cup \phi)} = \frac{P(HT)}{P(HT) + P(TH)}$$

$$= \frac{p(1-q)}{p(1-q) + (1-p)q}$$

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What is the expected payoff of a ticket (in dollars)?

$E(X) = \sum x P(X=x)$

$= \sum P(A_i) E[X|A_i]$

$\{A_1, A_2 \dots A_n\}$ forms partition of sample set.

$E[X] = \sum P(A_i) E[X|A_i]$

$E[X|A_i] = \sum x P(X=x|A_i)$

all are pairwise disjoint

$\bigcup_{i=1}^{n} A_n = \Omega$

initial Ticket

$T_1 \to \$10$

$T_1 \to \$1$

$T_1 \to F$ (2 free tickets)

$T_1 \to \$0$

Ticket is worth of 10 dollars

$\omega_1 : T_1 \to 10$

$\omega_2 : T_1 \to 1$

$\omega_3 : T_1 \to 0$ $\quad X(\omega_3) = 0$

$\omega_4 : T_1 \to F$, 2 free tickets. $\quad X(\omega_3) = 11$

$T_{11} \to 10$

$T_{12} \to 1$

Toss a coin $\quad \Omega = \{H, T\}$

Throw a dice $\quad \Omega = \{1, 2, 3, 4, 5, 6\}$

$\omega_5 : T_1 \to F$ $\quad T_{11}, T_{12}$

$T_{11} \to F$ $\quad T_{111} \to \$10$, $T_{112} \to \$5$

$T_{12} \to F$

$T_{121} \to 0$ $\quad T_{122} \to 10$

X: total dollar payoff generated starting initial ticket T_1

let $A_{10} = \{T_1 \text{ gives } \$10\}$

$A_1 = \{T_1 \text{ gives } \$1\}$

$A_F = \{T_1 \text{ gives 2 free tickets } T_{11}, T_{12}\}$

$A_0 = \{T_1 \text{ gives nothing}\}$

$P(A_{10}) = 1/20 \quad P(A_1) = 1/5 \quad P(A_F) = 1/4 \quad P(A_0) = 1/2$

$E[X] = \sum P(B_i) E[X|B_i]$

$= P(A_{10}) \underbrace{E[X|A_{10}]}_{10} + P(A_1) \underbrace{E[X|A_1]}_{1 \times 1} + P(A_F) \underbrace{E[X|A_F]}_{2E[X]} + P(A_0) \underbrace{E[X|A_0]}_{1 \times 0}$

$\frac{E(X)}{2} = \frac{1}{2} + \frac{1}{5} = \frac{7}{10}$

$E[X] = \frac{1}{20} \times 10 + \frac{1}{5} \times 1 + \frac{1}{4} \times 2E[X] + \frac{1}{2} \times 0$

$E[X] = 7/5 = 1.4$

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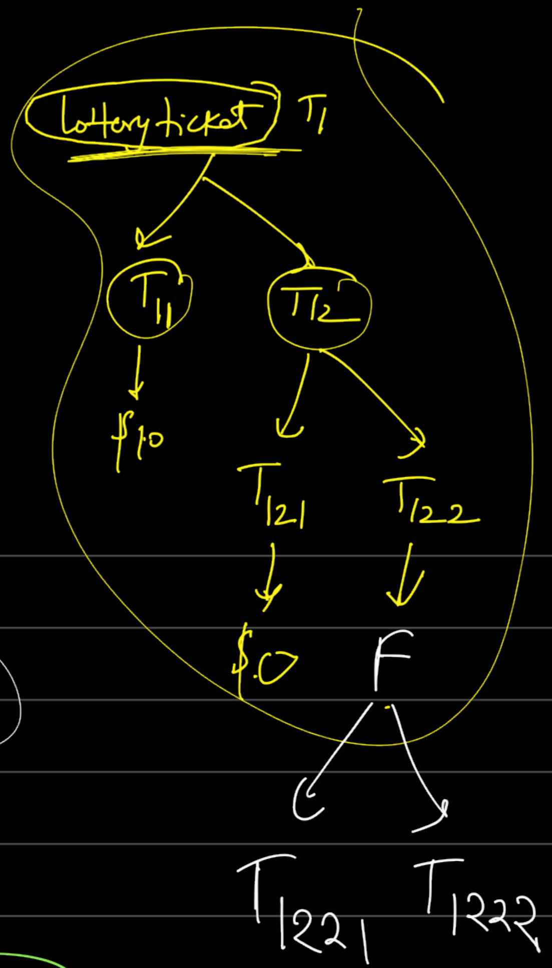

Partition of a set:

$\{A_1, A_2, A_3 \cdots A_{10}\}$

$\{\{1\}, \{2,3\}, \{4,5\}\}$

$E(X) = \sum P(A_i) E(X|A_i)$

law of expectation.

$E(X)$

Total probab: $P(X) = \sum P(A_i) P(X|A_i)$

$\boxed{E[X] = \sum x P(X=x)}$

$= \sum P(A_i) E[X|A_i]$

$E[X|A_i] = \sum x P(X=x|A_i)$

$= 10 \times 1 = 10$

use this

$E[X|A_{10}]$

$X : \to 10$

$P(X=10|A_{10}) = 1$

$P(A)$

$= P(A|\Omega)$

$P(A|B) \to$ Given B

$B \subseteq \Omega$

$E[X|A] = \sum x P(X=x|A) = 10 * 1 = 10$

$A_{10} \to T_1$ produces \$10.

X can takes 10

$E[X|A_F]$

always

$A_F = \{$ Ticket 1 gives u two tickets $T_{11}, T_{12} \}$

X_{11} = total pay-off generated from T_{11}

X_{12} = " " " " " T_{12}

$T_1 \to T_{11}, T_{12}$

$X = X_{11} + X_{12}$

$E[aX+bY] = aE[X] + bE[Y]$

$E[(aX+bY)|C] = aE[X|C] + bE[Y|C]$

$E[(X_{11}+X_{12})|A_F]$

$= E[X_{11}|A_F] + E[X_{12}|A_F]$

T_1, T_{11}, T_{12}

New ticket

" $E[X]$ $\qquad$ $E[X]$

T_1 (F)

Total sum

$T_{11} + T_{12}$

$E(X+Y) = E(X) + E(Y)$